

HDOC DAY -13

QUESTION :

Count the number of even digits, and odd digits. E.g. Number: 12045, Even_dig = 3, Odd_dig = 2

Algorithm

1. Start.
2. Input a positive integer n.
3. Display menu:
 - 1: Count even and odd digits
 - 2: Find sum of even and odd digits
4. Input user choice.
5. Use switch-case.
6. Extract each digit using $\text{digit} = n \% 10$.
7. Check whether digit is even or odd using $\text{digit} \% 2$.
8. Perform the selected operation.
9. Display result.
10. Stop.

C PROGRAM:

```
#include <stdio.h>

int main() {
    int n, temp, digit, choice;
    int evenCount = 0, oddCount = 0;
    int evenSum = 0, oddSum = 0;

    printf("Enter a positive integer: ");
```

```
scanf("%d", &n);
```

```
if (n <= 0) {  
    printf("Invalid input! Enter a positive integer.\n");  
    return 0;  
}
```

```
printf("\nMenu\n");  
printf("1. Count even and odd digits\n");  
printf("2. Find sum of even and odd digits\n");  
printf("Enter your choice: ");  
scanf("%d", &choice);
```

```
temp = n;
```

```
switch (choice) {  
    case 1:  
        while (temp > 0) {  
            digit = temp % 10;  
  
            if (digit % 2 == 0)  
                evenCount++;  
            else  
                oddCount++;  
  
            temp = temp / 10;  
        }  
}
```

```
printf("Even_dig = %d\n", evenCount);  
printf("Odd_dig = %d\n", oddCount);  
break;
```

case 2:

```
while (temp > 0) {  
    digit = temp % 10;  
  
    if (digit % 2 == 0)  
        evenSum = evenSum + digit;  
    else  
        oddSum = oddSum + digit;
```

```
    temp = temp / 10;  
}
```

```
printf("Sum_even = %d\n", evenSum);  
printf("Sum_odd = %d\n", oddSum);  
break;
```

default:

```
    printf("Invalid choice!\n");  
}
```

```
return 0;  
}
```

```

#include <stdio.h>

int main() {
    int n, temp, digit, choice;
    int evenCount = 0, oddCount = 0;
    int evenSum = 0, oddSum = 0;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    if (n <= 0) {
        printf("Invalid input! Enter a positive integer.\n");
        return 0;
    }

    printf("\nMenu\n");
    printf("1. Count even and odd digits\n");
    printf("2. Find sum of even and odd digits\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    temp = n;

    switch (choice) {
        case 1:
            while (temp > 0) {
                digit = temp % 10;

                if (digit % 2 == 0)
                    evenCount++;
                else
                    oddCount++;

                temp = temp / 10;
            }

            printf("Even_dig = %d\n", evenCount);
            printf("Odd_dig = %d\n", oddCount);
            break;

        case 2:
            while (temp > 0) {
                digit = temp % 10;

                if (digit % 2 == 0)
                    evenSum = evenSum + digit;
                else
                    oddSum = oddSum + digit;

                temp = temp / 10;
            }

            printf("Sum_even = %d\n", evenSum);
            printf("Sum_odd = %d\n", oddSum);
            break;

        default:
            printf("Invalid choice!\n");
    }

    return 0;
}

```

COMPILATION AND EXECUTION:

```
(kali㉿kali)-[~]
└─$ vi h13.c

(kali㉿kali)-[~]
└─$ gcc h13.c -o h13

(kali㉿kali)-[~]
└─$ ./h13
Enter a positive integer: 12045

Menu
1. Count even and odd digits
2. Find sum of even and odd digits
Enter your choice: 1
Even_dig = 3
Odd_dig = 2
```

TEST CASE TABLE:

TEST CASE	NUMBER	CHOICE	EXPECTED OUTPUT
1	12045	1	Even digits= 3, odd digits = 2
2	12345	2	Sum of even digits=6, sum of odd digits=9
3	86420	1	Even digits=5, odd digits=0
4	11111	2	Sum of even digits=0, sum of odd digits=5
5	90909	1	Even digits= 2, odd digits=3

```
(kali㉿kali) [~]
$ ./h13
Enter a positive integer: 12045

Menu
1. Count even and odd digits
2. Find sum of even and odd digits
Enter your choice: 1
Even_dig = 3
Odd_dig = 2

(kali㉿kali)-[~]
$ ./h13
Enter a positive integer: 12345

Menu
1. Count even and odd digits
2. Find sum of even and odd digits
Enter your choice: 2
Sum_even = 6
Sum_odd = 9

(kali㉿kali)-[~]
$ ./h13
Enter a positive integer: 86420

Menu
1. Count even and odd digits
2. Find sum of even and odd digits
Enter your choice: 1
Even_dig = 5
Odd_dig = 0

(kali㉿kali)-[~]
$ ./h13
Enter a positive integer: 11111

Menu
1. Count even and odd digits
2. Find sum of even and odd digits
Enter your choice: 2
Sum_even = 0
Sum_odd = 5

(kali㉿kali)-[~]
$ ./h13
Enter a positive integer: 90909

Menu
1. Count even and odd digits
2. Find sum of even and odd digits
```